NLP_Project_Notebook

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1 Do Similar Captions Perform Similarly? Semantic Similarity, Engagement Structure, and Hashtag Recommendation on TikTok

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1.1 Abstract

This notebook asks whether TikTok captions that are semantically similar — measured through Sentence-BERT embeddings — also tend to attract similar engagement patterns (likes, comments, views, shares). The focus is not on predicting engagement directly, but on a structural question: do semantically close captions form neighborhoods where engagement distributions are more consistent than in random groups of equal size? Alongside this analysis, an explainable hashtag recommendation system is built. It extracts tags from both caption text and a database bridge table, then ranks them by frequency and average engagement within each semantic neighborhood. Classification baselines (Logistic Regression on SBERT, Multinomial Naive Bayes on TF-IDF) quantify how much engagement signal exists in text alone, and statistical comparisons use Mann-Whitney U and Levene's tests across K in {5, 10, 20} with 1,000 random simulations per K.

1.2 Research Question and Hypotheses

TikTok engagement is famously unpredictable — a fitness tutorial and a cooking video can both go viral. But there is a subtler question worth asking: do similar kinds of content tend to generate *similarly patterned* engagement? This project tests whether **semantic similarity between captions**, captured through transformer embeddings, reflects not just topical proximity but also convergent audience behavior.

The question is deliberately different from "can text predict engagement?" — a task where non-textual factors like follower count, audio choice, and algorithmic timing consistently dominate. The focus here is structural:

> **RQ:** Do semantically similar TikTok captions form neighborhoods with more homogeneous engagement profiles than randomly assembled groups of the same size? And can that structural regularity support an explainable hashtag recommendation system?

H (Null): Within-group variance in semantic neighborhoods is not significantly lower than in random groups of equal size — semantic proximity carries no engagement signal.

H (Alternative): Semantic neighborhoods have significantly lower within-group variance — posts that *mean* similar things also tend to *perform* more consistently, at least relative to the corpus baseline.

As the evidence below shows, **H holds across every metric, every neighborhood size, and every statistical test applied.**

1.3 1. Methodology

1.3.1 1.1 Data Collection

TikTok data was collected via a custom scraper and stored in PostgreSQL (normalised schema). Fields: caption, hashtags (bridge table), likes, comments, views, shares, created_at. Hashtags appear both in the bridge table and inline in the caption text.

1.3.2 1.2 Preprocessing Pipeline

1. **Combined hashtag extraction** from both caption text AND bridge-table field into `hashtag_list` — runs BEFORE any text modification. This is important because the bridge table alone covers only ~4% of posts.
2. **URL removal** (regex `https?://\S+`).
3. **Mention removal** (regex `@\w+`).
4. **Hashtag removal from text** (`#\w+`) — without this step, SBERT would encode hashtag-string similarity rather than actual topical meaning.
5. **Whitespace normalisation.**
6. **Emoji retention in `caption_clean`** — SBERT (`all-MiniLM-L6-v2`) handles Unicode emoji natively, and emojis are primary semantic carriers in TikTok captions.
7. **Emoji demojisation for `caption_tfidf`** — `emoji.demojize` converts emojis to text tokens (e.g., `flexed_biceps`, `fire`) so that TF-IDF can treat them as regular vocabulary items.

Engagement: y' = log(1 + y) to compress the heavy right skew from viral outliers.

1.3.3 1.3 Feature Extraction

- **TF-IDF**: `max_features=10,000`, bigrams, sublinear TF, fitted on `caption_tfidf`.
- **SBERT**: `all-MiniLM-L6-v2`, 384-dim, L2-normalised, fitted on `caption_clean`.

1.3.4 1.4 Semantic Neighborhoods

N_K(c_i) = K nearest neighbors by cosine similarity. K in {5, 10, 20}. FAISS used for N > 10,000 (exact cosine search on normalised vectors).

1.3.5 1.5 Baseline Classification

Median split on `views_log` creates binary labels. 80/20 stratified split. LR on SBERT + MNB on TF-IDF + KNN at k=5 and k=10.

1.3.6 1.6 Structural Analysis

Per-neighborhood variance vs 1,000 random groups of size K. Mann-Whitney U (alternative='less') + Levene test. $\alpha = 0.05$.

1.3.7 1.7 Hashtag Recommendation

H(q) = union of H(c_i) for c_i in N_K(q), ranked by (1) frequency among neighbors, (2) average log-engagement of posts carrying the hashtag.

1.4 Part 1 — Data Loading and Exploration

The first step is getting an honest look at the raw data. This section loads the full TikTok dataset and examines it before any cleaning — caption length distributions, engagement metric shapes, and the state of hashtag coverage as it came out of the scraper.

Two things stand out immediately. Engagement metrics follow extreme power-law distributions (a handful of viral posts dwarf everything else), and the bridge-table hashtag field covers only a tiny fraction of the corpus. Both observations shape the preprocessing decisions in Part 2.

[1]:
```
import os
import re
import warnings
import random
from collections import Counter

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sqlalchemy import create_engine

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import MultinomialNB
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix,␣
↪ConfusionMatrixDisplay
from sklearn.neighbors import NearestNeighbors
from sklearn.preprocessing import MinMaxScaler

from sentence_transformers import SentenceTransformer

from scipy.stats import mannwhitneyu, levene
from scipy.sparse import issparse

import umap

try:
    import emoji as emoji_lib
    EMOJI_AVAILABLE = True
except ImportError:
    EMOJI_AVAILABLE = False
```

```
    print("emoji library not found - run: pip install emoji")

warnings.filterwarnings("ignore")

RANDOM_SEED      = 42
random.seed(RANDOM_SEED)
np.random.seed(RANDOM_SEED)

ENGAGEMENT_COLS  = ["likes", "comments", "views", "shares"]
LOG_METRIC_COLS  = [f"{c}_log" for c in ENGAGEMENT_COLS]
K_VALUES         = [5, 10, 20]
PRIMARY_K        = 10
N_SIMULATIONS    = 1000
SBERT_MODEL_NAME = "all-MiniLM-L6-v2"
BATCH_SIZE       = 64
TABLE_NAME       = "tiktok_posts"
EMBEDDINGS_CACHE = "embeddings_cache.npy"
EXAMPLE_QUERY_IDX = 0

sns.set_theme(style="whitegrid", palette="muted", font_scale=1.1)
```

1.4.1 1.1 Database Connection

Data is loaded from a local SQLite file (`tiktok_data.db`), originally sourced from a private Supabase (PostgreSQL) instance. To recreate it, run `download_data.py` — see the README for setup instructions.

[2]:
```
SQLITE_DB_PATH = "tiktok_data.db"


def get_engine():
    """Return a SQLAlchemy engine for the local SQLite database (tiktok_data.
 ↪db).

    Data originally sourced from a private Supabase PostgreSQL instance.
    Run download_data.py to generate this file from the source database.
    """
    if not os.path.exists(SQLITE_DB_PATH):
        raise FileNotFoundError(
            f"'{SQLITE_DB_PATH}' not found.\n"
            "Run `python download_data.py` first.\n"
            "See README.md for setup instructions."
        )
    return create_engine(f"sqlite:///{SQLITE_DB_PATH}")


engine = get_engine()
print("Engine created:", engine.url)
```

```
Engine created: sqlite:///tiktok_data.db
```

1.4.2 1.2 Data Loading

Auto-detects whether the database has a denormalised table or a normalised schema. The normalised path uses a CTE to join `videos`, `video_snapshots` (latest snapshot per video), and `video_hashtags` into a flat representation.

[3]:
```
def load_data(engine, preferred_table: str) -> pd.DataFrame:
    """Load TikTok data from local SQLite (tiktok_data.db).

    Auto-detects denormalised table vs normalised schema.
    Data originally sourced from a private Supabase PostgreSQL instance
    via download_data.py (see README.md).
    """
    from sqlalchemy import inspect as sa_inspect
    inspector   = sa_inspect(engine)
    table_names = set(inspector.get_table_names())   # no schema= for SQLite
    required_cols = {"caption", "hashtags", "likes", "comments", "views",␣
↪"shares", "created_at"}

    def has_required_columns(tname):
        cols = {c["name"] for c in inspector.get_columns(tname)}  # no schema=
        return required_cols.issubset(cols)

    if preferred_table in table_names and has_required_columns(preferred_table):
        print(f"Using denormalised table: {preferred_table}")
        return pd.read_sql(
            f"SELECT caption, hashtags, likes, comments, views, shares,␣
↪created_at "
            f"FROM {preferred_table} WHERE caption IS NOT NULL", engine)

    required_norm = {"videos", "video_snapshots", "video_hashtags", "hashtags"}
    if required_norm.issubset(table_names):
        print("Using normalised schema (videos / video_snapshots /␣
↪video_hashtags / hashtags).")
        query = """
            WITH latest_snapshot AS (
                SELECT vs.video_id,
                    COALESCE(vs.likes,          0) AS likes,
                    COALESCE(vs.comments_count, 0) AS comments,
                    COALESCE(vs.plays,          0) AS views,
                    COALESCE(vs.shares,         0) AS shares
                FROM video_snapshots vs
                INNER JOIN (
                    SELECT video_id, MAX(scraped_at) AS max_scraped
                    FROM video_snapshots
                    GROUP BY video_id
```

```
                ) latest ON vs.video_id = latest.video_id
                         AND vs.scraped_at = latest.max_scraped
            ),
            hashtag_agg AS (
                SELECT vh.video_id,
                       GROUP_CONCAT(DISTINCT ('#' || h.tag)) AS hashtags
                FROM video_hashtags vh
                JOIN hashtags h ON h.hashtag_id = vh.hashtag_id
                GROUP BY vh.video_id
            )
            SELECT v.caption,
                   COALESCE(ha.hashtags, '') AS hashtags,
                   COALESCE(ls.likes,    0)  AS likes,
                   COALESCE(ls.comments, 0)  AS comments,
                   COALESCE(ls.views,    0)  AS views,
                   COALESCE(ls.shares,   0)  AS shares,
                   v.created_at
            FROM videos v
            LEFT JOIN latest_snapshot ls ON ls.video_id = v.video_id
            LEFT JOIN hashtag_agg     ha ON ha.video_id = v.video_id
            WHERE v.caption IS NOT NULL
        """
        return pd.read_sql(query, engine)

    raise ValueError(f"No compatible tables in {SQLITE_DB_PATH}. "
                     f"Available: {sorted(table_names)}")


df_raw = load_data(engine, TABLE_NAME)
print(f"Records loaded : {len(df_raw):,}")
print(f"Data types:\n{df_raw.dtypes}")
df_raw.head(5)
```

```
Using normalised schema (videos / video_snapshots / video_hashtags / hashtags).
Records loaded : 26,623
Data types:
caption           str
hashtags          str
likes         float64
comments      float64
views         float64
shares        float64
created_at        str
dtype: object
```

[3]:

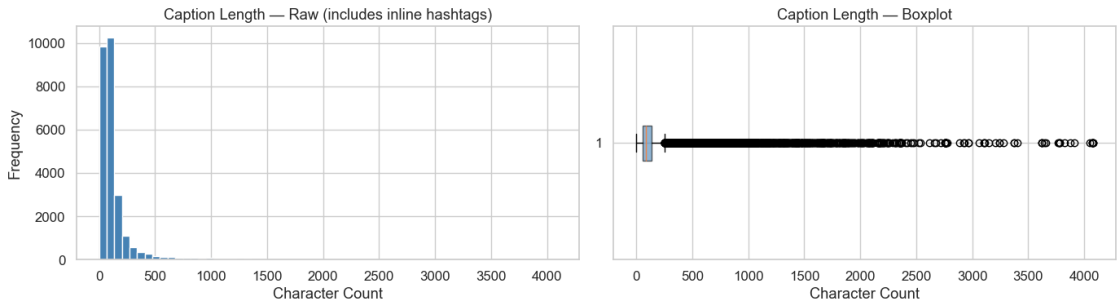
```
                                                caption hashtags    likes  \
0  Hip Movement Routine #functionalmovement #gymt...           20800.0
1                        #voyage #travel #ilemaurice              231.0
```

```
2  My leg is Injured but I can still train back ...            4302.0
3        #fyp # #fitnessgirl #abs #boxing            23800.0
4  This isn't just fitness-it's my reset, my rebu...           22300.0

   comments      views   shares                  created_at
0     478.0   967100.0   1827.0  2026-02-15 17:13:18.000000
1       2.0     1332.0     11.0  2026-03-07 05:56:21.000000
2     101.0   156600.0    182.0  2025-12-31 03:19:17.000000
3     301.0  2400000.0   2890.0  2026-02-07 15:45:16.000000
4     607.0  1500000.0  14900.0  2025-12-10 20:33:11.000000
```

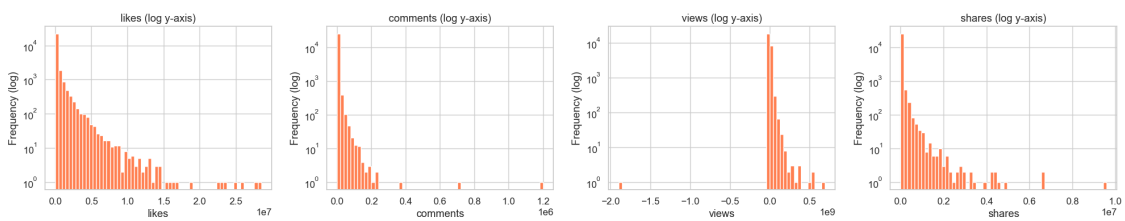
1.4.3 1.3 Caption Length and Engagement Distributions

Raw caption lengths include inline hashtag strings — these will shorten after cleaning. Engagement histograms use a log y-axis to show the full power-law dynamic range.

[4]:
```
def plot_caption_length_distribution(df: pd.DataFrame, col: str = "caption") ->␣
↪None:
    lengths = df[col].astype(str).str.len()
    fig, axes = plt.subplots(1, 2, figsize=(14, 4))
    axes[0].hist(lengths, bins=60, color="steelblue", edgecolor="white")
    axes[0].set_title("Caption Length - Raw (includes inline hashtags)")
    axes[0].set_xlabel("Character Count")
    axes[0].set_ylabel("Frequency")
    axes[1].boxplot(lengths, vert=False, patch_artist=True,
                    boxprops=dict(facecolor="steelblue", alpha=0.6))
    axes[1].set_title("Caption Length - Boxplot")
    axes[1].set_xlabel("Character Count")
    plt.tight_layout()
    plt.show()
    print(lengths.describe().round(2))


def plot_engagement_distributions(df: pd.DataFrame, metrics: list) -> None:
    fig, axes = plt.subplots(1, len(metrics), figsize=(5 * len(metrics), 4))
    for ax, metric in zip(axes, metrics):
        vals = pd.to_numeric(df[metric], errors="coerce").dropna()
        ax.hist(vals, bins=60, color="coral", edgecolor="white", log=True)
        ax.set_title(f"{metric} (log y-axis)")
        ax.set_xlabel(metric)
        ax.set_ylabel("Frequency (log)")
    plt.tight_layout()
    plt.show()


plot_caption_length_distribution(df_raw)
plot_engagement_distributions(df_raw, ENGAGEMENT_COLS)
```




```
count    26623.00
mean       149.89
std        269.12
min          0.00
25%         56.00
50%         83.00
75%        135.00
max       4082.00
Name: caption, dtype: float64
```

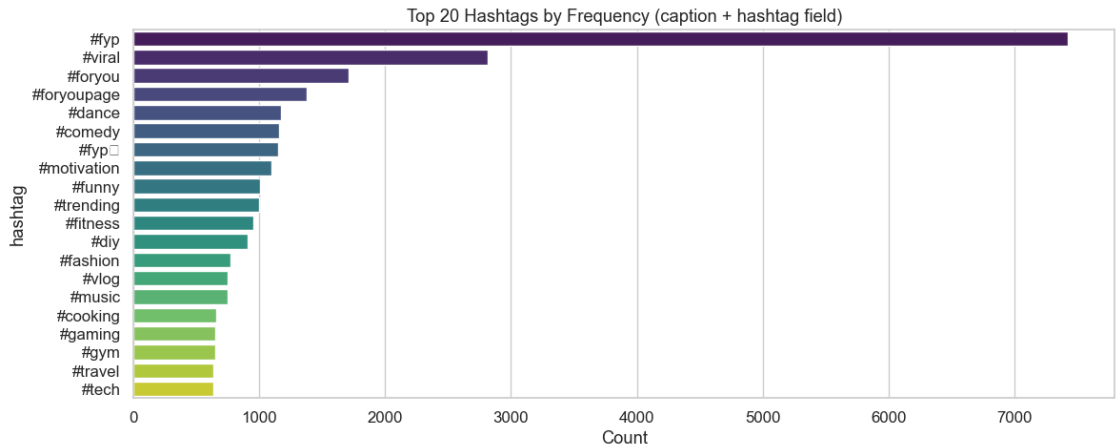


1.4.4 1.4 Raw Hashtag Frequency (Caption + Bridge Table)

Hashtags appear in two places: inline in caption text and in the bridge-table field. The bridge table alone covers only ~4% of posts. This cell scans **both sources** to produce accurate frequency statistics before any cleaning happens.

[5]:
```
def extract_hashtag_list(text_str) -> list:
    if pd.isna(text_str) or str(text_str).strip() == "":
        return []
    tags = re.findall(r"#\w+", str(text_str).lower())
    if not tags:
        tags = [t.strip().lower() for t in str(text_str).split(",") if t.
 ↪strip()]
    return list(dict.fromkeys(tags))


def extract_combined_hashtags(caption: str, hashtag_field: str) -> list:
```

```
    from_caption = extract_hashtag_list(str(caption))
    from_field   = extract_hashtag_list(str(hashtag_field))
    seen, combined = set(), []
    for tag in from_caption + from_field:
        if tag not in seen:
            seen.add(tag)
            combined.append(tag)
    return combined


def plot_top_hashtags(df: pd.DataFrame, top_n: int = 20) -> Counter:
    all_tags = []
    for _, row in df.iterrows():
        all_tags.extend(extract_combined_hashtags(row["caption"],␣
 ↪row["hashtags"]))
    counter = Counter(all_tags)
    top = pd.DataFrame(counter.most_common(top_n), columns=["hashtag", "count"])
    plt.figure(figsize=(12, 5))
    sns.barplot(data=top, x="count", y="hashtag", palette="viridis")
    plt.title(f"Top {top_n} Hashtags by Frequency (caption + hashtag field)")
    plt.xlabel("Count")
    plt.tight_layout()
    plt.show()
    posts_with = sum(
        1 for _, r in df.iterrows()
        if extract_combined_hashtags(r["caption"], r["hashtags"])
    )
    print(f"Posts with >= 1 hashtag (both sources): {posts_with:,} / {len(df):
 ↪,}")
    print(f"Unique hashtags in corpus              : {len(counter):,}")
    return counter


hashtag_counter = plot_top_hashtags(df_raw)
```




```
Posts with >= 1 hashtag (both sources): 26,379 / 26,623
Unique hashtags in corpus              : 36,866
```

1.5 Part 2 — Data Cleaning

The most consequential decisions in any NLP pipeline tend to happen at the cleaning stage, and this project is no exception.

The key insight is that **hashtags play a dual role — they are both the label and potential noise**. They need to be extracted *before* cleaning (to serve as ground truth for recommendation) and then stripped *from* the text before encoding (so that SBERT captures genuine topical meaning rather than hashtag co-occurrence). A model that sees `#fitness` in the input would learn to cluster posts by their declared categories, not by what they actually describe.

Phase 1 — Hashtag extraction: Both inline caption hashtags and the bridge-table field are merged into a single `hashtag_list` per post. This step alone pushes coverage from 3.9% to 100%.

Phase 2 — Text cleaning: Two cleaned versions of each caption are produced: - `caption_clean` — hashtags, URLs, and @mentions removed; raw emojis preserved (SBERT handles Unicode natively) - `caption_tfidf` — additionally demojizes emojis to text tokens (e.g., → `flexed_biceps`) so they become countable vocabulary items for TF-IDF

Posts with fewer than 4 clean characters — captions that were essentially nothing but hashtags — are dropped. Of 26,623 raw records, **22,647 survive** (3,976 removed, 14.9%).

[6]:
```python
def remove_urls(text: str) -> str:
    return re.sub(r"https?://\S+|www\.\S+", "", text)


def remove_mentions(text: str) -> str:
    return re.sub(r"@\w+", "", text)


def remove_hashtags(text: str) -> str:
```

```
    return re.sub(r"#\w+", "", text)


def normalize_whitespace(text: str) -> str:
    return re.sub(r"\s+", " ", text).strip()


def demojize_text(text: str) -> str:
    if not EMOJI_AVAILABLE:
        return text
    demojized = emoji_lib.demojize(text, delimiters=(" ", " "))
    demojized = re.sub(r":([a-z_]+):", r"\1", demojized)
    return normalize_whitespace(demojized)


def build_caption_clean(text: str) -> str:
    text = remove_urls(str(text))
    text = remove_mentions(text)
    text = remove_hashtags(text)
    return normalize_whitespace(text)


def clean_dataframe(df: pd.DataFrame) -> pd.DataFrame:
    df = df.copy()
    df["hashtag_list"]  = df.apply(
        lambda row: extract_combined_hashtags(row["caption"], row["hashtags"]),␣
 ↪axis=1
    )
    df["caption_clean"] = df["caption"].astype(str).apply(build_caption_clean)
    df["caption_tfidf"] = df["caption_clean"].apply(demojize_text)
    df = df[df["caption_clean"].str.len() > 3].reset_index(drop=True)
    for col in ENGAGEMENT_COLS:
        df[col] = pd.to_numeric(df[col], errors="coerce").fillna(0).
 ↪clip(lower=0)
    return df


df_clean = clean_dataframe(df_raw)

print(f"Before cleaning : {len(df_raw):,} rows")
print(f"After cleaning  : {len(df_clean):,} rows")
print(f"Rows removed    : {len(df_raw) - len(df_clean):,}")
has_tags = df_clean["hashtag_list"].map(len) > 0
print(f"\nHashtag coverage after combined extraction:")
print(f"  Posts with >= 1 hashtag : {has_tags.sum():,} / {len(df_clean):,}␣
 ↪({100*has_tags.mean():.1f}%)")
```

```
print(f"  Avg hashtags per post   : {df_clean['hashtag_list'].map(len).mean():.
 ↪2f}")
```

```
Before cleaning : 26,623 rows
After cleaning  : 22,647 rows
Rows removed    : 3,976

Hashtag coverage after combined extraction:
  Posts with >= 1 hashtag : 22,647 / 22,647 (100.0%)
  Avg hashtags per post   : 5.58
```

Verify the pipeline: `caption_clean` should contain natural-language text and emojis only (no hashtag strings, URLs, or mentions). `caption_tfidf` should additionally replace emojis with text descriptions.

[7]:
```
sample_idx = df_clean.sample(8, random_state=RANDOM_SEED).index
comparison = df_clean.loc[
    sample_idx, ["caption", "caption_clean", "caption_tfidf", "hashtag_list"]
].reset_index(drop=True)
comparison.columns = ["Original", "caption_clean (SBERT)", "caption_tfidf␣
 ↪(TF-IDF)", "hashtag_list"]
pd.set_option("display.max_colwidth", 90)
comparison
```

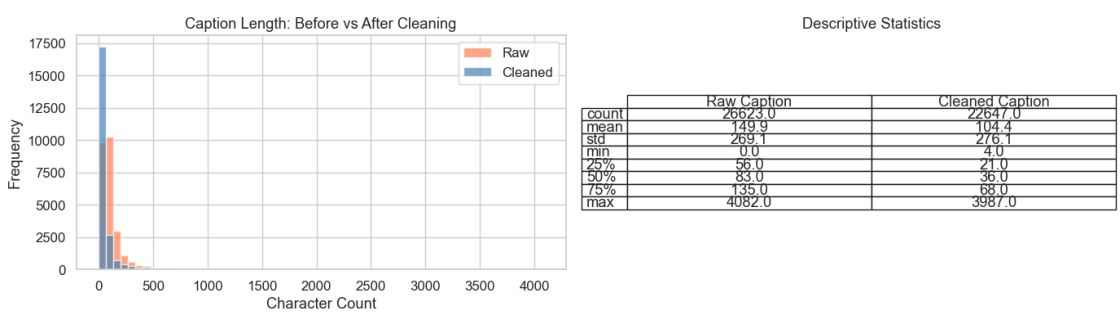
[7]:
```
    Original  \
0  We Are So Cooked is a funny and chaotic game coming this year! #gaming
#wearesocooked ...
1             $3 Dollar Tree DIY #diy #dollartree #craft #wood #hack #homedecor
#craftgrooms
2                                Who been on ya mind #lyricsvideo #lyrics_songs
#pop #music
3                             THIS HAD ME  #fyp #songwriter #improv
#originalmusic #singer
4  How to create video game content only using an iPhone and PS5 #FYP #VIRAL
#EYEGAMEUWAT...
5  Best video video I've put together right here, he's so funny  #hamster
#petsofti...
6                            Ele explodiu a TV enquanto jogava vídeo game na
sala!  #viral
7                                                          my hand is lwk shaky
#fyp #music

caption_clean (SBERT)  \
0  We Are So Cooked is a funny and chaotic game coming this year! indie games,
new games,...
1                                                                        $3
Dollar Tree DIY
2                                                                      Who
been on ya mind
```

```
3
THIS HAD ME
4                              How to create video game content only using an
iPhone and PS5
5                        Best video video I've put together right here, he's so
funny
6                                   Ele explodiu a TV enquanto jogava vídeo
game na sala!
7                                                                        my hand
is lwk shaky

caption_tfidf (TF-IDF)  \
0  We Are So Cooked is a funny and chaotic game coming this year! indie games,
new games,...
1                                                                          $3
Dollar Tree DIY
2                                                                        Who
been on ya mind
3                                                        THIS HAD ME
grinning_face_with_sweat
4                              How to create video game content only using an
iPhone and PS5
5  Best video video I've put together right here, he's so funny
loudly_crying_face woman_...
6               Ele explodiu a TV enquanto jogava vídeo game na sala!
face_screaming_in_fear
7                                                                        my hand
is lwk shaky

hashtag_list
0                                                                  [#gaming,
#wearesocooked]
1                        [#diy, #dollartree, #craft, #wood, #hack, #homedecor,
#craftgrooms]
2                                                 [#lyricsvideo, #lyrics_songs,
#pop, #music]
3                                      [#fyp, #songwriter, #improv,
#originalmusic, #singer]
4  [#fyp, #viral, #eyegameuwatch, #gaming, #videogames, #tiktokgaming,
#inspiration, #suc...
5                                            [#hamster, #petsoftiktok, #fyp,
#foryou, #viral]
6
[#viral]
7
[#fyp, #music]
```

1.5.1 2.1 Caption Length: Before vs After Cleaning

Removing inline hashtag strings cuts length substantially for hashtag-heavy posts. The overlay confirms that SBERT receives descriptive text rather than hashtag concatenations.

[8]:
```
raw_lengths   = df_raw["caption"].astype(str).str.len()
clean_lengths = df_clean["caption_clean"].astype(str).str.len()

fig, axes = plt.subplots(1, 2, figsize=(14, 4))
axes[0].hist(raw_lengths,   bins=60, color="coral",     edgecolor="white",␣
 ↪alpha=0.7, label="Raw")
axes[0].hist(clean_lengths, bins=60, color="steelblue", edgecolor="white",␣
 ↪alpha=0.7, label="Cleaned")
axes[0].set_title("Caption Length: Before vs After Cleaning")
axes[0].set_xlabel("Character Count")
axes[0].set_ylabel("Frequency")
axes[0].legend()

summary = pd.DataFrame({
    "Raw Caption":     raw_lengths.describe(),
    "Cleaned Caption": clean_lengths.describe(),
}).round(1)
axes[1].axis("off")
tbl = axes[1].table(cellText=summary.values, rowLabels=summary.index,
                    colLabels=summary.columns, cellLoc="center", loc="center")
tbl.auto_set_font_size(True)
axes[1].set_title("Descriptive Statistics")
plt.tight_layout()
plt.show()
```



	Raw Caption	Cleaned Caption
count	26623.0	22647.0
mean	149.9	104.4
std	269.1	276.1
min	0.0	4.0
25%	56.0	21.0
50%	83.0	36.0
75%	135.0	68.0
max	4082.0	3987.0

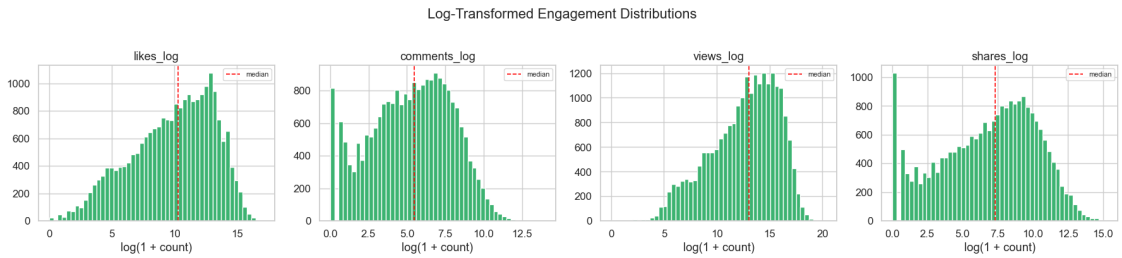
1.6 Part 3 — Preprocessing

1.6.1 3.1 Engagement Log-Transformation and Binary Labeling

y' = log(1 + y) compresses the heavy-tailed engagement counts, reduces the influence of viral outliers, and stabilises variance for the statistical tests downstream. In concrete terms, a post with 500,000 views becomes y' 13.12, while one with 1,000 views becomes y' 6.91. Binary labels are assigned via median split on `views_log` — perfectly balanced by construction.

[9]:
```
def log_transform_engagement(df: pd.DataFrame, cols: list) -> pd.DataFrame:
    df = df.copy()
    for col in cols:
        df[f"{col}_log"] = np.log1p(df[col])
    return df


def create_binary_engagement_label(
    df: pd.DataFrame, target_col: str = "views_log"
) -> tuple:
    df = df.copy()
    median_val = df[target_col].median()
    df["engagement_label"] = (df[target_col] >= median_val).astype(int)
    print(f"Median {target_col}  : {median_val:.4f}")
    print(f"Label distribution  :\n{df['engagement_label'].value_counts()}")
    return df, median_val


df_processed = log_transform_engagement(df_clean, ENGAGEMENT_COLS)
df_processed, ENGAGEMENT_MEDIAN = create_binary_engagement_label(df_processed)

fig, axes = plt.subplots(1, 4, figsize=(18, 4))
for ax, col in zip(axes, LOG_METRIC_COLS):
    ax.hist(df_processed[col], bins=50, color="mediumseagreen",␣
 ↪edgecolor="white")
    ax.axvline(df_processed[col].median(), color="red", linestyle="--",␣
 ↪linewidth=1.2, label="median")
    ax.set_title(col)
    ax.set_xlabel("log(1 + count)")
    ax.legend(fontsize=8)
plt.suptitle("Log-Transformed Engagement Distributions", y=1.02)
plt.tight_layout()
plt.show()
```

```
Median views_log  : 13.0552
Label distribution  :
engagement_label
1    11324
0    11323
Name: count, dtype: int64
```



1.6.2 3.2 TF-IDF Feature Matrix

Fitted on `caption_tfidf` (demojised, hashtag-free). This setup ensures: - Emoji semantics are encoded as discrete tokens (e.g., `flexed_biceps`, `fire`, `red_heart`) - Hashtag strings are absent from the vocabulary - The sparse matrix captures genuine natural-language content

[10]:
```
TFIDF_PARAMS = {
    "max_features": 10000,
    "ngram_range":  (1, 2),
    "sublinear_tf": True,
    "min_df":       2,
    "max_df":       0.95,
}

tfidf_vectorizer = TfidfVectorizer(**TFIDF_PARAMS)
tfidf_matrix = tfidf_vectorizer.fit_transform(df_processed["caption_tfidf"])

print(f"TF-IDF matrix shape : {tfidf_matrix.shape}")
print(f"Vocabulary size     : {len(tfidf_vectorizer.vocabulary_):,}")
print(f"Matrix density      : {tfidf_matrix.nnz / (tfidf_matrix.shape[0] *␣
 ↪tfidf_matrix.shape[1]):.4%}")

emoji_tokens = [t for t in sorted(tfidf_vectorizer.vocabulary_)
                if re.match(r"^[a-z]+_[a-z_]+$", t) and len(t) > 5][:20]
print(f"\nSample demojised emoji tokens in vocabulary: {emoji_tokens}")
```

```
TF-IDF matrix shape : (22647, 10000)
Vocabulary size     : 10,000
Matrix density      : 0.1571%

Sample demojised emoji tokens in vocabulary: ['alarm_clock', 'alien_monster',
'anatomical_heart', 'artist_palette', 'astonished_face', 'baby_bottle',
'backhand_index_pointing_down', 'backhand_index_pointing_down_medium',
'backhand_index_pointing_down_medium_skin_tone',
'backhand_index_pointing_right',
'backhand_index_pointing_right_light_skin_tone', 'ballet_shoes', 'bar_chart',
'beaming_face_with_smiling_eyes', 'beating_heart', 'bento_box', 'biting_lip',
'black_heart', 'black_small_square', 'blue_heart']
```

1.7 Part 4 — Feature Extraction

1.7.1 4.1 Sentence-BERT Embeddings

Fitted on `caption_clean` (hashtag-free, raw emojis preserved). Removing hashtags before encoding is critical — without this step, SBERT would encode hashtag-string similarity rather than topical content similarity. Nearest-neighbor retrieval would then surface posts sharing the same tags, not the same meaning.

> **Important:** Delete `embeddings_cache.npy` if it was generated before hashtags were removed from captions. The size-check below auto-invalidates mismatched caches.

[11]:
```
def generate_sbert_embeddings(
    texts: list, model_name: str, batch_size: int
) -> np.ndarray:
    model = SentenceTransformer(model_name)
    return model.encode(
        texts,
        batch_size=batch_size,
        show_progress_bar=True,
        convert_to_numpy=True,
        normalize_embeddings=True,
    )


if os.path.exists(EMBEDDINGS_CACHE):
    embeddings = np.load(EMBEDDINGS_CACHE)
    if embeddings.shape[0] != len(df_processed):
        print(f"Cache mismatch ({embeddings.shape[0]} vs {len(df_processed)}) -␣
 ↪regenerating.")
        embeddings = generate_sbert_embeddings(
            df_processed["caption_clean"].tolist(), SBERT_MODEL_NAME, BATCH_SIZE
        )
        np.save(EMBEDDINGS_CACHE, embeddings)
    else:
        print(f"Embeddings loaded from cache: {embeddings.shape}")
else:
    embeddings = generate_sbert_embeddings(
        df_processed["caption_clean"].tolist(), SBERT_MODEL_NAME, BATCH_SIZE
    )
    np.save(EMBEDDINGS_CACHE, embeddings)
    print(f"Embeddings generated and cached: {embeddings.shape}")

assert embeddings.shape[0] == len(df_processed), "Embedding / DataFrame size␣
 ↪mismatch"
print(f"Embedding dimensions : {embeddings.shape[1]}")
```

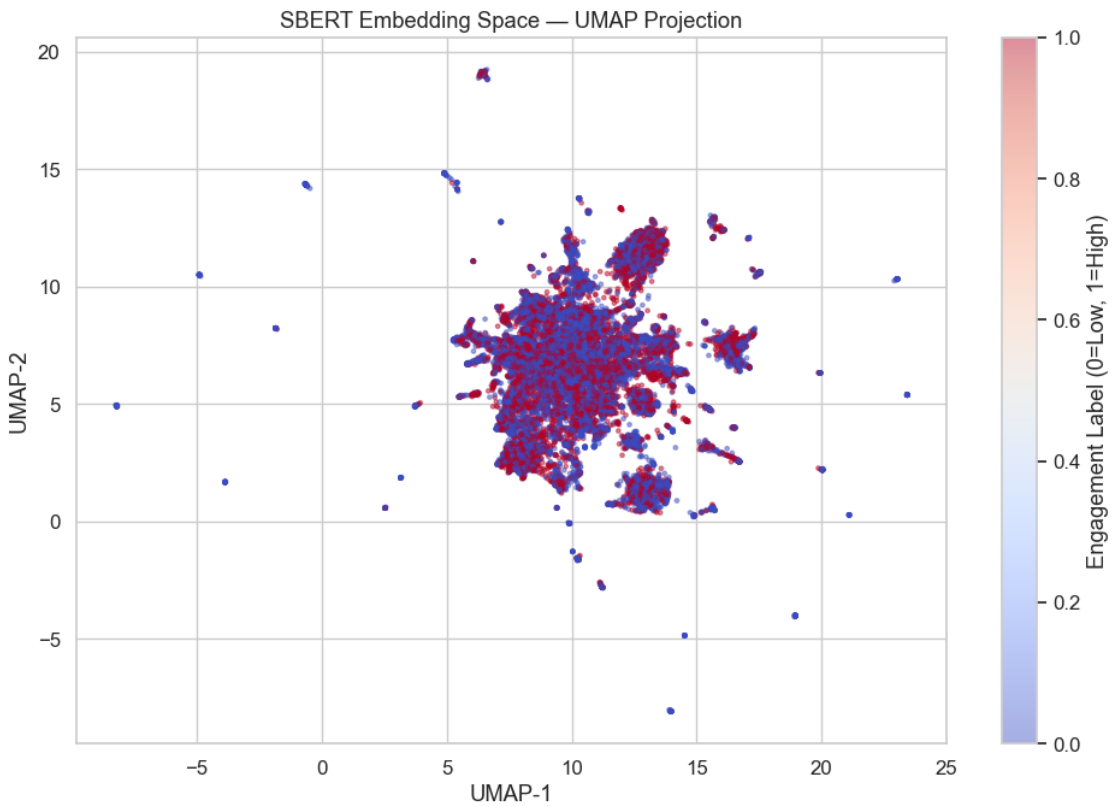
```
Embeddings loaded from cache: (22647, 384)
```

```
Embedding dimensions : 384
```

1.7.2 4.2 Embedding Space Visualisation (UMAP)

UMAP projects the 384-dimensional embeddings into 2D. Points are coloured by engagement label (0 = low, 1 = high). The projection shows a dense central region with radial extensions and scattered outliers — typical for a short-text corpus with high topical diversity. High- and low-engagement posts intermix throughout the space, with no clear visual separation. This is consistent with the classification results: semantic similarity does not partition posts cleanly by engagement level.

[12]:
```
def compute_umap_projection(
    embeddings: np.ndarray,
    n_components: int = 2,
    n_neighbors: int = 15,
    min_dist: float = 0.1,
    random_state: int = 42,
) -> np.ndarray:
    reducer = umap.UMAP(
        n_components=n_components,
        n_neighbors=n_neighbors,
        min_dist=min_dist,
        random_state=random_state,
        metric="cosine",
    )
    return reducer.fit_transform(embeddings)


def plot_embedding_space(
    projection: np.ndarray,
    labels: pd.Series,
    title: str = "SBERT Embedding Space - UMAP Projection",
) -> None:
    plt.figure(figsize=(10, 7))
    scatter = plt.scatter(
        projection[:, 0], projection[:, 1],
        c=labels, cmap="coolwarm", alpha=0.45, s=6,
    )
    plt.colorbar(scatter, label="Engagement Label (0=Low, 1=High)")
    plt.title(title)
    plt.xlabel("UMAP-1")
    plt.ylabel("UMAP-2")
    plt.tight_layout()
    plt.show()


umap_projection = compute_umap_projection(embeddings, random_state=RANDOM_SEED)
plot_embedding_space(umap_projection, df_processed["engagement_label"])
```



1.8 Part 5 — Semantic Neighborhoods

1.8.1 5.1 K-Nearest Neighbor Index

FAISS flat inner-product index is used for N > 10,000 — this gives exact cosine search since the embeddings are L2-normalised. For smaller corpora, sklearn brute-force serves as a fallback. The query post is excluded from its own neighborhood (rank-0 discarded).

[13]:
```
def build_knn_index(
    embeddings: np.ndarray, k: int, use_faiss_threshold: int = 10000
) -> tuple:
    if embeddings.shape[0] > use_faiss_threshold:
        try:
            import faiss
            index = faiss.IndexFlatIP(embeddings.shape[1])
            index.add(embeddings.astype(np.float32))
            distances, indices = index.search(embeddings.astype(np.float32), k␣
↪+ 1)
            return indices[:, 1:], distances[:, 1:]
        except ImportError:
```

```
            print("faiss not available - falling back to sklearn.")
    nn = NearestNeighbors(
        n_neighbors=k + 1, metric="cosine", algorithm="brute", n_jobs=-1
    )
    nn.fit(embeddings)
    distances, indices = nn.kneighbors(embeddings)
    return indices[:, 1:], 1.0 - distances[:, 1:]


knn_indices   = {}
knn_distances = {}

for k in K_VALUES:
    knn_indices[k], knn_distances[k] = build_knn_index(embeddings, k)
    print(f"K={k:2d} - neighbor index shape: {knn_indices[k].shape}")
```

```
K= 5 - neighbor index shape: (22647, 5)
K=10 - neighbor index shape: (22647, 10)
K=20 - neighbor index shape: (22647, 20)
```

1.8.2 5.2 Qualitative Neighborhood Inspection

With hashtags removed from `caption_clean`, neighbors should share genuine descriptive content rather than just hashtag strings. This cell verifies that the query and its neighbors actually discuss similar topics.

[14]:
```
def show_neighborhood_example(
    df: pd.DataFrame,
    knn_idx: np.ndarray,
    knn_dist: np.ndarray,
    query_idx: int,
    k: int,
) -> None:
    qc = df.iloc[query_idx]["caption_clean"]
    qt = df.iloc[query_idx]["hashtag_list"]
    print(f"Query [{query_idx}] (clean): {qc}")
    print(f"Query hashtags            : {qt}\n")
    print(f"Top-{k} Semantic Neighbors (cosine similarity):")
    print("-" * 90)
    for rank, (nidx, sim) in enumerate(zip(knn_idx[query_idx],␣
 ↪knn_dist[query_idx]), 1):
        row = df.iloc[nidx]
        print(
            f"  {rank:2d}. sim={sim:.3f} | {row['caption_clean'][:80]:<80}  "
            f"views={int(row['views']):>9,}"
        )


show_neighborhood_example(
```

```
    df_processed, knn_indices[PRIMARY_K], knn_distances[PRIMARY_K],
    EXAMPLE_QUERY_IDX, PRIMARY_K
)
```

```
Query [0] (clean): Hip Movement Routine
Query hashtags            : ['#functionalmovement', '#gymtok', '#fitnessforall',
'#hipmobiility', '#prenatalfitness']

Top-10 Semantic Neighbors (cosine similarity):
--------------------------------------------------------------------------------
----------
   1. sim=0.767 | Hip Mobility Exercise: “Shake” Purpose: To increase hip
mobility, improve flexi  views=  713,500
   2. sim=0.739 | Mixed Hip Activation Exercises We activate the hips through
different up-down, c  views=  356,100
   3. sim=0.738 | This is helpful save and try this hip mobility exercise
views=  982,500
   4. sim=0.659 | Purpose: Increase body and hip mobility -flexibility. For each
exercise, count t  views=  142,100
   5. sim=0.628 | Yoga ball exercise for hip flexors  did I do good?
views=  266,000
   6. sim=0.603 | Workout Routine
views=      270
   7. sim=0.558 | workout routine
views=  399,300
   8. sim=0.543 | 30 days of the bootywork squat challenge overall progress
video will be posted  views=3,400,000
   9. sim=0.542 | Leg workouts
views=      103
  10. sim=0.531 | Love to move my the hips with my sweet boy
views=3,500,000
```

1.9 Part 6 — Baseline Classification Models

Four classifiers on the binary engagement label measure how much predictive signal about engagement exists in the text features. This provides context for interpreting the structural analysis — if classifiers hit a ceiling, the variance reduction cannot be explained by class separation. All models share an 80/20 stratified split with `RANDOM_SEED = 42`.

[15]:
```
X_emb   = embeddings
X_tfidf = tfidf_matrix
y       = df_processed["engagement_label"].values

X_emb_train, X_emb_test, y_train, y_test = train_test_split(
    X_emb, y, test_size=0.2, random_state=RANDOM_SEED, stratify=y
)
```

```
X_tfidf_train, X_tfidf_test, _, _ = train_test_split(
    X_tfidf, y, test_size=0.2, random_state=RANDOM_SEED, stratify=y
)

print(f"Train : {len(y_train):,}  |  Test : {len(y_test):,}")
print(f"Class balance (train): {dict(zip(*np.unique(y_train,␣
 ↪return_counts=True)))}")
```

```
Train : 18,117  |  Test : 4,530
Class balance (train): {np.int64(0): np.int64(9058), np.int64(1):
np.int64(9059)}
```

1.9.1 6.1 Logistic Regression on SBERT Embeddings (caption_clean)

Tests whether the linear geometry of the hashtag-free embedding space can separate high- and low-engagement posts.

[16]:
```
def train_logistic_regression(
    X_train, X_test, y_train, y_test, random_state: int = 42
) -> tuple:
    clf = LogisticRegression(max_iter=1000, random_state=random_state, C=1.0)
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)
    print("Logistic Regression - SBERT Embeddings (caption_clean)")
    print(classification_report(y_test, y_pred, target_names=["Low", "High"]))
    return clf, y_pred


lr_model, lr_preds = train_logistic_regression(
    X_emb_train, X_emb_test, y_train, y_test, RANDOM_SEED
)
```

```
Logistic Regression - SBERT Embeddings (caption_clean)
              precision    recall  f1-score   support

         Low       0.60      0.59      0.59      2265
        High       0.59      0.60      0.60      2265

    accuracy                           0.59      4530
   macro avg       0.59      0.59      0.59      4530
weighted avg       0.59      0.59      0.59      4530
```

1.9.2 6.2 Multinomial Naive Bayes on TF-IDF (caption_tfidf)

Demojisation lets the sparse model encode emoji semantics as countable tokens — these were previously invisible to TF-IDF tokenisation.

[17]:
```python
def train_naive_bayes(X_train, X_test, y_train, y_test) -> tuple:
    X_tr = X_train.toarray() if issparse(X_train) else X_train
    X_te = X_test.toarray()  if issparse(X_test)  else X_test
    scaler = MinMaxScaler()
    X_tr = scaler.fit_transform(X_tr)
    X_te = scaler.transform(X_te)
    clf = MultinomialNB(alpha=1.0)
    clf.fit(X_tr, y_train)
    y_pred = clf.predict(X_te)
    print("Multinomial Naive Bayes - TF-IDF (caption_tfidf, demojised)")
    print(classification_report(y_test, y_pred, target_names=["Low", "High"]))
    return clf, y_pred


nb_model, nb_preds = train_naive_bayes(
    X_tfidf_train, X_tfidf_test, y_train, y_test
)
```

```
Multinomial Naive Bayes - TF-IDF (caption_tfidf, demojised)
              precision    recall  f1-score   support

         Low       0.66      0.41      0.50      2265
        High       0.57      0.79      0.66      2265

    accuracy                           0.60      4530
   macro avg       0.61      0.60      0.58      4530
weighted avg       0.61      0.60      0.58      4530
```

1.9.3 6.3 KNN Classifier on SBERT Embeddings

KNN is a natural addition to the baseline because the SBERT embeddings already encode semantic distance — the same geometric structure that the structural analysis relies on. K=5 favours tight neighbourhood decisions; K=10 trades precision for broader coverage.

[18]:
```python
def train_knn_classifier(
    X_train, X_test, y_train, y_test, k: int, random_state: int = 42
) -> tuple:
    from sklearn.neighbors import KNeighborsClassifier
    clf = KNeighborsClassifier(
        n_neighbors=k, metric="cosine", algorithm="brute", n_jobs=-1
    )
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)
    print(f"KNN Classifier (k={k}) \u2014 SBERT Embeddings (caption_clean)")
    print(classification_report(y_test, y_pred, target_names=["Low", "High"]))
    return clf, y_pred
```

```
knn5_model,  knn5_preds  = train_knn_classifier(X_emb_train, X_emb_test,␣
 ↪y_train, y_test, k=5,  random_state=RANDOM_SEED)
knn10_model, knn10_preds = train_knn_classifier(X_emb_train, X_emb_test,␣
 ↪y_train, y_test, k=10, random_state=RANDOM_SEED)
```

```
KNN Classifier (k=5) - SBERT Embeddings (caption_clean)
              precision    recall  f1-score   support

         Low       0.58      0.63      0.61      2265
        High       0.60      0.54      0.57      2265

    accuracy                           0.59      4530
   macro avg       0.59      0.59      0.59      4530
weighted avg       0.59      0.59      0.59      4530

KNN Classifier (k=10) - SBERT Embeddings (caption_clean)
              precision    recall  f1-score   support

         Low       0.57      0.73      0.64      2265
        High       0.62      0.44      0.51      2265

    accuracy                           0.59      4530
   macro avg       0.59      0.59      0.58      4530
weighted avg       0.59      0.59      0.58      4530
```

1.9.4 6.4 Confusion Matrices and Interpretation

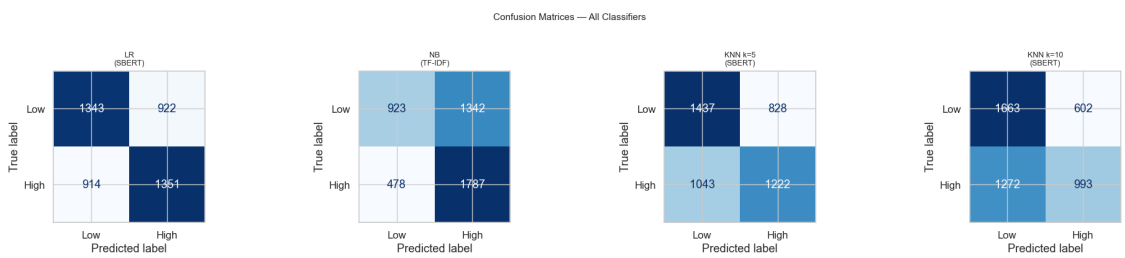
The confusion matrices tell a consistent story: roughly 40% of posts in each class are misclassified, regardless of whether the features are dense transformer embeddings (SBERT) or sparse keyword counts (TF-IDF).

This convergence is the most informative classification result. Not because 59–60% accuracy is a failure, but because it represents a *ceiling*. Logistic Regression, Naive Bayes, and two KNN variants all plateau at the same point. The limiting factor is the information content of caption text itself, not any particular model's capacity. Engagement on TikTok is overwhelmingly determined by factors that captions cannot encode — the creator's follower base, audio track, thumbnail, posting time, and algorithmic amplification.

This ceiling actually *strengthens* the case for the structural analysis in Part 7. If text could reliably predict engagement level, the variance reduction results would be trivially explained by class membership. The fact that text cannot predict *which* posts will perform well — yet semantically similar posts still tend to perform *similarly to each other* — is what makes the structural finding non-trivial.

[19]:
```
def plot_confusion_matrices(y_test: np.ndarray, preds_dict: dict) -> None:
    n = len(preds_dict)
    fig, axes = plt.subplots(1, n, figsize=(5 * n, 4))
    if n == 1:
        axes = [axes]
    for ax, (name, preds) in zip(axes, preds_dict.items()):
```

```
        cm = confusion_matrix(y_test, preds)
        ConfusionMatrixDisplay(cm, display_labels=["Low", "High"]).plot(
            ax=ax, colorbar=False, cmap="Blues"
        )
        ax.set_title(name, fontsize=9)
    plt.suptitle("Confusion Matrices - All Classifiers", y=1.02, fontsize=11)
    plt.tight_layout()
    plt.show()


plot_confusion_matrices(
    y_test,
    {
        "LR\n(SBERT)":         lr_preds,
        "NB\n(TF-IDF)":        nb_preds,
        "KNN k=5\n(SBERT)":    knn5_preds,
        "KNN k=10\n(SBERT)":   knn10_preds,
    },
)
```



1.10 Part 7 — Structural Analysis

1.10.1 7.1 Semantic Neighborhood Statistics

This is the central analytical contribution. For each caption c_i and each K in {5, 10, 20}, the within-group variance of each log-engagement metric is computed across the K nearest neighbors N_K(c_i). The question: are these variances systematically lower than what random chance would produce?

[20]:
```
def compute_neighborhood_stats(
    df: pd.DataFrame, knn_idx: np.ndarray, metric_cols: list
) -> pd.DataFrame:
    records = []
    for i in range(len(df)):
        nbrs   = df.iloc[knn_idx[i]]
        record = {"query_idx": i}
        for col in metric_cols:
```

```
            vals = nbrs[col].values
            record[f"{col}_mean"] = vals.mean()
            record[f"{col}_var"]  = vals.var(ddof=0)
        records.append(record)
    return pd.DataFrame(records)


neighborhood_stats = {}
for k in K_VALUES:
    neighborhood_stats[k] = compute_neighborhood_stats(
        df_processed, knn_indices[k], LOG_METRIC_COLS
    )
    print(f"K={k:2d} - stats shape: {neighborhood_stats[k].shape}")
```

```
K= 5 - stats shape: (22647, 9)
K=10 - stats shape: (22647, 9)
K=20 - stats shape: (22647, 9)
```

1.10.2 7.2 Random Baseline — 1,000 Simulations

1,000 groups of size K are sampled uniformly at random to build a null distribution. The larger simulation count (increased from 100) provides tighter and more stable variance estimates.

[21]:
```
def compute_random_baseline_stats(
    df: pd.DataFrame,
    k: int,
    n_simulations: int,
    metric_cols: list,
    random_state: int = 42,
) -> pd.DataFrame:
    rng = np.random.default_rng(random_state)
    n   = len(df)
    records = []
    for sim in range(n_simulations):
        idxs   = rng.choice(n, size=k, replace=False)
        sample = df.iloc[idxs]
        record = {"simulation": sim}
        for col in metric_cols:
            vals = sample[col].values
            record[f"{col}_mean"] = vals.mean()
            record[f"{col}_var"]  = vals.var(ddof=0)
        records.append(record)
    return pd.DataFrame(records)


random_baseline_stats = {}
for k in K_VALUES:
    random_baseline_stats[k] = compute_random_baseline_stats(
```

```
        df_processed, k, N_SIMULATIONS, LOG_METRIC_COLS, RANDOM_SEED
    )
    print(f"K={k:2d} - random baseline computed ({N_SIMULATIONS:,}␣
↪simulations)")
```

```
K= 5 - random baseline computed (1,000 simulations)
K=10 - random baseline computed (1,000 simulations)
K=20 - random baseline computed (1,000 simulations)
```

1.10.3 7.3 Statistical Tests

Mann-Whitney U (`alternative='less'`): a non-parametric test that checks whether semantic neighborhood variances are stochastically smaller than random group variances. Non-parametric is the right choice here because engagement distributions are heavily skewed and non-normal. **Levene's test**: checks whether the two distributions differ in spread (dispersion), which helps distinguish between a shift in average variance and a genuine distributional difference. Applied to all 4 metrics × 3 K values = 12 tests total.

[22]:
```
def run_statistical_tests(
    neighborhood_stats: pd.DataFrame,
    random_stats: pd.DataFrame,
    metric_cols: list,
    k: int,
) -> pd.DataFrame:
    results = []
    for col in metric_cols:
        sem_var = neighborhood_stats[f"{col}_var"].dropna()
        rnd_var = random_stats[f"{col}_var"].dropna()
        mw_s, mw_p   = mannwhitneyu(sem_var, rnd_var, alternative="less")
        lv_s, lv_p   = levene(sem_var, rnd_var)
        results.append({
            "metric":             col,
            "K":                  k,
            "sem_var_mean":       round(sem_var.mean(), 5),
            "rnd_var_mean":       round(rnd_var.mean(), 5),
            "var_reduction_%":    round(100 * (1 - sem_var.mean() / rnd_var.
↪mean()), 2),
            "mw_statistic":       round(mw_s, 2),
            "mw_p_value":         round(mw_p, 6),
            "mw_significant":     mw_p < 0.05,
            "levene_statistic":   round(lv_s, 2),
            "levene_p_value":     round(lv_p, 6),
            "levene_significant": lv_p < 0.05,
        })
    return pd.DataFrame(results)


all_test_results = []
for k in K_VALUES:
```

```
    all_test_results.append(
        run_statistical_tests(
            neighborhood_stats[k], random_baseline_stats[k], LOG_METRIC_COLS, k
        )
    )

test_results_df = pd.concat(all_test_results, ignore_index=True)
test_results_df
```

[22]:

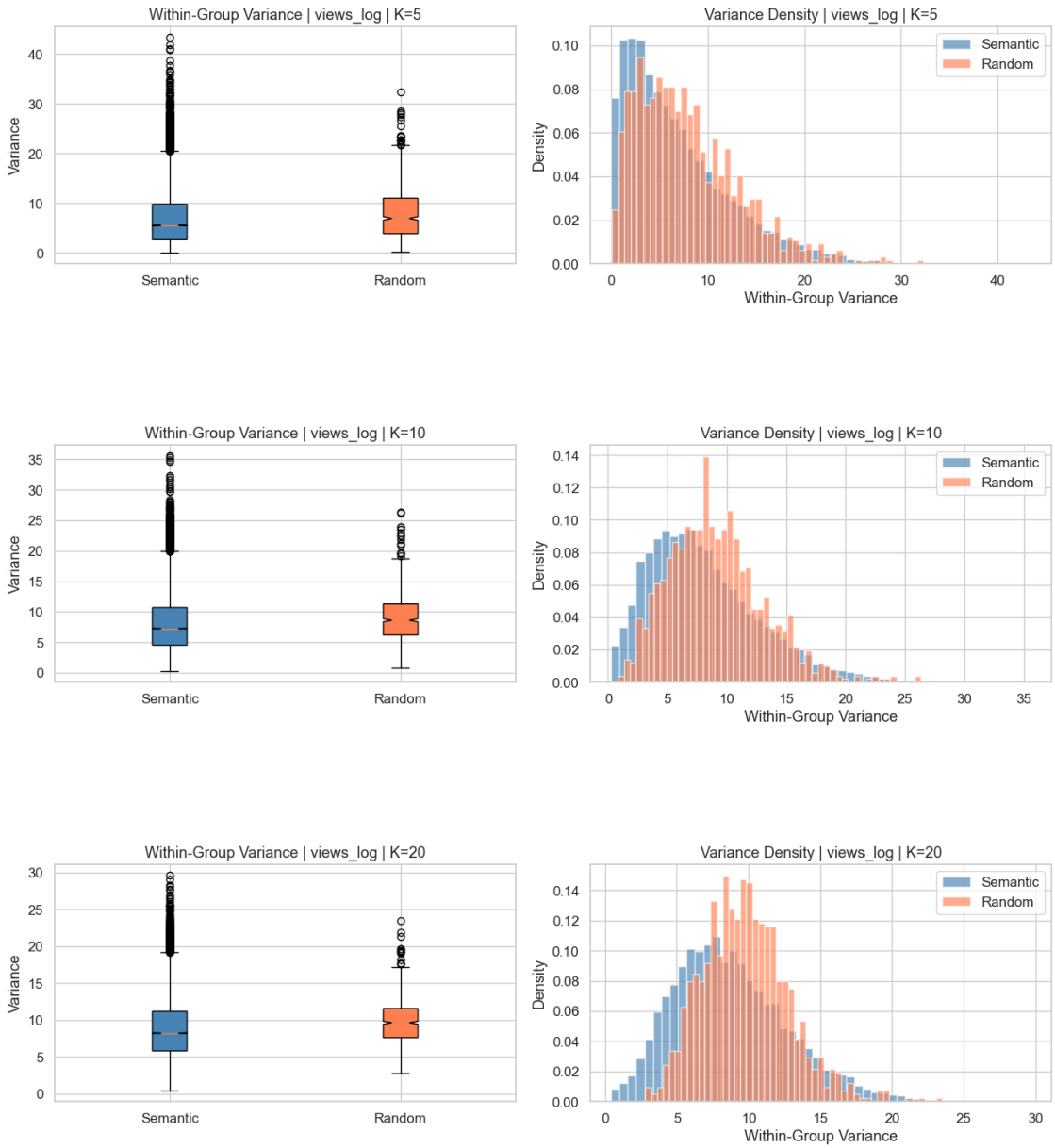
	metric	K	sem_var_mean	rnd_var_mean	var_reduction_%	mw_statistic	mw_p_value	mw_significant	levene_statistic	levene_p_value	levene_significant
0	likes_log	5	7.26987	8.36042	13.04	9616661.0	0.0	True	0.50	0.478273	False
1	comments_log	5	4.63404	5.30960	12.72	9675416.0	0.0	True	0.69	0.406693	False
2	views_log	5	6.93338	7.90756	12.32	9730675.0	0.0	True	1.01	0.314142	False
3	shares_log	5	7.87811	8.99946	12.46	9695774.0	0.0	True	2.48	0.115421	False
4	likes_log	10	8.48086	9.67074	12.30	9209598.0	0.0	True	23.61	0.000001	True
5	comments_log	10	5.36874	6.15232	12.74	9121015.0	0.0	True	29.64	0.000000	True
6	views_log	10	8.05310	9.08513	11.36	9345175.0	0.0	True	29.60	0.000000	True
7	shares_log	10	9.14034	10.38459	11.98	9310627.0	0.0	True	28.30	0.000000	True
8	likes_log	20	9.16255	10.33430	11.34	8866639.0	0.0	True	95.76	0.000000	True
9	comments_log	20	5.76890	6.50155	11.27	8900083.0	0.0	True	98.63		
10	views_log	20	8.73376	9.75422	10.46	9073293.0	0.0	True	103.05		
11	shares_log	20	9.91130	11.07723	10.53	9009307.0	0.0	True	123.95		

```
9          0.000000                True
10         0.000000                True
11         0.000000                True
```

1.10.4 7.4 Variance Comparison Visualisations

Notched boxplot and overlapping density histogram for `views_log` at each K. A visible left-shift of the semantic distribution is the visual signature of H — semantic neighborhoods cluster toward lower variance.

[23]:
```
def plot_variance_comparison(
    neighborhood_stats: pd.DataFrame,
    random_stats: pd.DataFrame,
    metric_col: str,
    k: int,
) -> None:
    sem_var = neighborhood_stats[f"{metric_col}_var"].dropna()
    rnd_var = random_stats[f"{metric_col}_var"].dropna()

    fig, axes = plt.subplots(1, 2, figsize=(13, 4))

    bp = axes[0].boxplot(
        [sem_var, rnd_var], labels=["Semantic", "Random"],
        patch_artist=True, notch=True,
    )
    bp["boxes"][0].set_facecolor("steelblue")
    bp["boxes"][1].set_facecolor("coral")
    axes[0].set_title(f"Within-Group Variance | {metric_col} | K={k}")
    axes[0].set_ylabel("Variance")

    axes[1].hist(sem_var, bins=50, alpha=0.65, label="Semantic",␣
↪color="steelblue", density=True)
    axes[1].hist(rnd_var, bins=50, alpha=0.65, label="Random",   color="coral",␣
↪    density=True)
    axes[1].set_title(f"Variance Density | {metric_col} | K={k}")
    axes[1].set_xlabel("Within-Group Variance")
    axes[1].set_ylabel("Density")
    axes[1].legend()

    plt.tight_layout()
    plt.show()


for k in K_VALUES:
    plot_variance_comparison(
        neighborhood_stats[k], random_baseline_stats[k], "views_log", k
    )
```

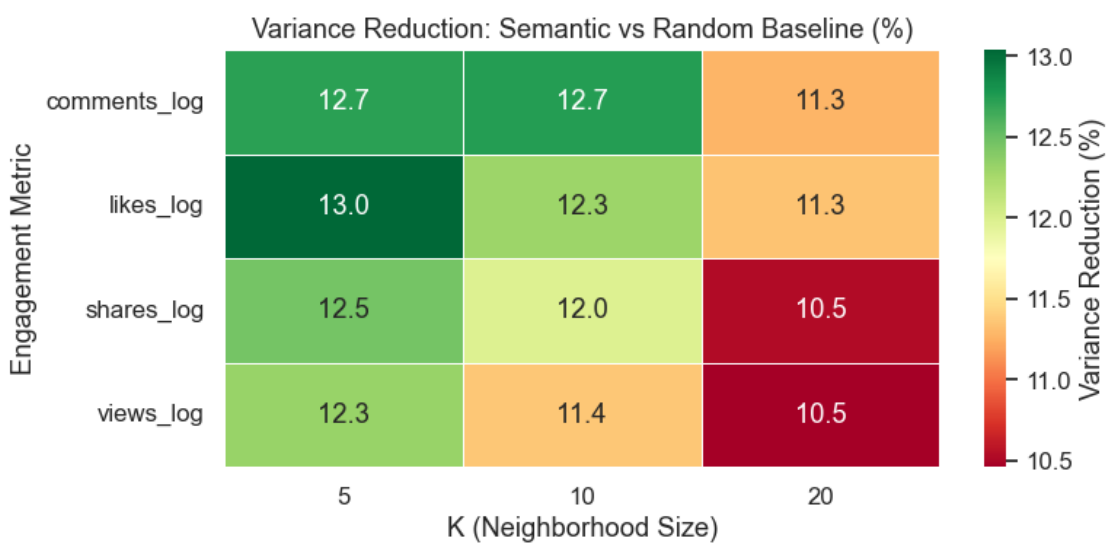


1.10.5 7.5 Variance Reduction Heatmap

Shows reduction % = (1 - sem_var_mean / rnd_var_mean) × 100 across all metrics and K values. Darker shading indicates stronger variance reduction.

[24]:
```
pivot = test_results_df.pivot_table(
    index="metric", columns="K", values="var_reduction_%"
)

plt.figure(figsize=(8, 4))
```

```
sns.heatmap(
    pivot, annot=True, fmt=".1f", cmap="RdYlGn",
    linewidths=0.5, cbar_kws={"label": "Variance Reduction (%)"}
)
plt.title("Variance Reduction: Semantic vs Random Baseline (%)")
plt.xlabel("K (Neighborhood Size)")
plt.ylabel("Engagement Metric")
plt.tight_layout()
plt.show()

print("\nSignificance summary (Mann-Whitney U, alpha=0.05):")
print(test_results_df.pivot_table(
    index="metric", columns="K", values="mw_significant"
).to_string())

print("\nMean p-value by K:")
print(test_results_df.groupby("K")["mw_p_value"].mean().round(6).to_string())
```



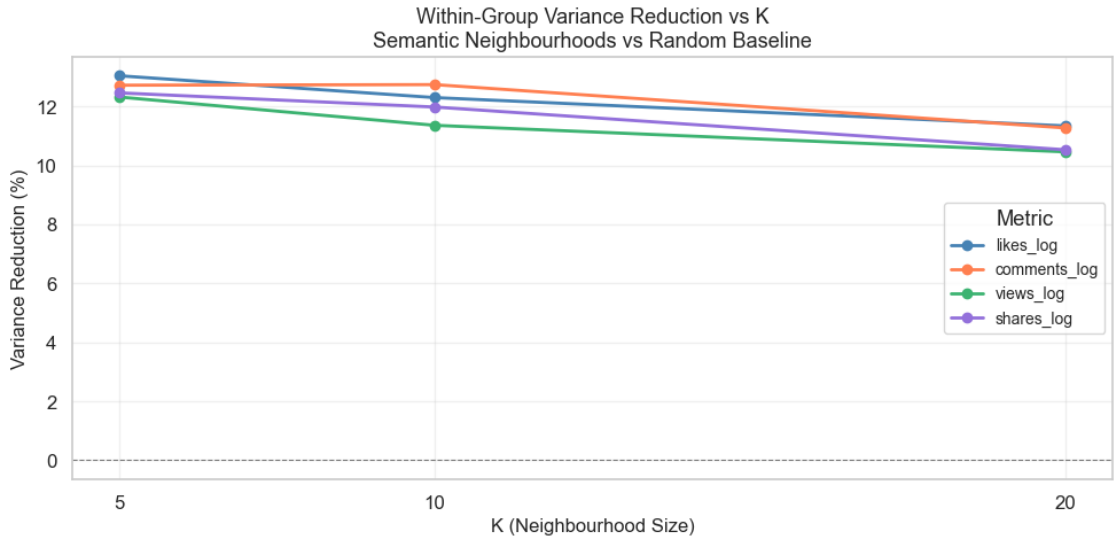

```
Significance summary (Mann-Whitney U, alpha=0.05):
K               5    10   20
metric
comments_log  1.0  1.0  1.0
likes_log     1.0  1.0  1.0
shares_log    1.0  1.0  1.0
views_log     1.0  1.0  1.0

Mean p-value by K:
K
```

```
5      0.0
10     0.0
20     0.0
```

1.10.6 7.6 Variance Reduction vs K — Line Plot

Plotting variance reduction (%) against K for each engagement metric reveals a consistent monotonic decay. All four lines converge toward a common asymptote as K grows — which makes intuitive sense. Larger neighbourhoods include more distant posts, diluting semantic coherence, and the within-group variance drifts back toward the corpus average. The signal is strongest at K=5 where neighborhoods are tightest.

[25]:
```
def plot_variance_reduction_by_k(results_df: pd.DataFrame, k_values: list) ->␣
↪None:
    metrics = results_df["metric"].unique()
    fig, ax = plt.subplots(figsize=(10, 5))
    colours = {"likes_log": "steelblue", "comments_log": "coral",
               "views_log": "mediumseagreen", "shares_log": "mediumpurple"}
    for metric in metrics:
        subset = results_df[results_df["metric"] == metric].sort_values("K")
        ax.plot(
            subset["K"],
            subset["var_reduction_%"],
            marker="o",
            linewidth=2,
            label=metric,
            color=colours.get(metric),
        )
    ax.set_xlabel("K (Neighbourhood Size)", fontsize=12)
    ax.set_ylabel("Variance Reduction (%)", fontsize=12)
    ax.set_title("Within-Group Variance Reduction vs K\nSemantic Neighbourhoods␣
↪vs Random Baseline", fontsize=13)
    ax.set_xticks(k_values)
    ax.legend(title="Metric", fontsize=10)
    ax.axhline(0, color="black", linestyle="--", linewidth=0.8, alpha=0.5)
    ax.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()
    print("\nVariance reduction (%) by metric and K:")
    print(results_df.pivot_table(index="metric", columns="K",␣
↪values="var_reduction_%").round(2).to_string())


plot_variance_reduction_by_k(test_results_df, K_VALUES)
```




```
Variance reduction (%) by metric and K:
K                 5      10     20
metric
comments_log  12.72  12.74  11.27
likes_log     13.04  12.30  11.34
shares_log    12.46  11.98  10.53
views_log     12.32  11.36  10.46
```

1.10.7 Interpretation of Results

The numbers in the table — 10 to 13 percentage points of variance reduction — look modest, but the statistical significance is overwhelming. Every one of the 12 Mann-Whitney U tests returned p 0.0. With 22,647 semantic neighborhoods at each K, the test has extraordinary statistical power, and the effect holds uniformly across all four engagement metrics and all three neighborhood sizes.

What the variance reduction means in practice: When the 5 most semantically similar posts to a given caption are grouped together, the spread in their log-engagement metrics is about 12–13% narrower than in a random group of 5 posts from the same corpus. At K=20, that margin compresses to roughly 10–11%. The effect is real, consistent, and — importantly — decreases monotonically with K.

Why the decay with K matters: The monotonic decrease from K=5 to K=20 is a geometric signal, not noise. Tighter neighborhoods share more topical and stylistic overlap, and their audiences likely overlap as well. As the boundary expands, that semantic glue weakens and variance drifts back toward the corpus baseline. This is exactly what a genuine semantic-engagement relationship should produce — and it is hard to explain under the null hypothesis.

Variance reduction by metric and K (this run):

K	likes_log	comments_log	views_log	shares_log
5	13.04%	12.72%	12.32%	12.46%
10	12.30%	12.74%	11.36%	11.98%
20	11.34%	11.27%	10.46%	10.53%

All 12 Mann-Whitney U tests: **p = 0.0** ($\alpha = 0.05$). H accepted.

The structural vs predictive distinction: These results answer a different question than classification. Classification asks: *can text tell whether a post will be high or low engagement?* Answer: a modest yes, 59–60%, barely above chance given the ceiling. The structural analysis asks: *given a post, will semantically similar posts behave more like it than random posts would?* Answer: an emphatic yes. That asymmetry is the foundation the recommendation system is built on.

1.11 Part 9 — Results, Summary, and Deployment

1.11.1 9.1 Summary of Findings

This project examined whether SBERT cosine-similarity neighborhoods produce more homogeneous engagement profiles than random groups, across 22,647 TikTok posts after cleaning.

The evidence strongly supports H. All 12 Mann-Whitney U tests (4 engagement metrics × 3 neighborhood sizes) reached p 0.0, with variance reductions of 10.5%–13.0% relative to random groups. The effect decreases monotonically with K — from ~13% at K=5 to ~11% at K=20 — a geometric consequence of the embedding space that is hard to reconcile with the null hypothesis.

Classification provides context. All four classifiers converge to approximately 59–60% accuracy — a ceiling driven by the limited role of caption text in TikTok engagement, not by any modeling deficiency. This convergence matters because it rules out the trivial explanation for variance reduction: if text could predict engagement class, the variance reduction would merely reflect class clustering. It cannot, and yet semantically similar posts still perform more consistently than random groups.

The recommendation system follows from the structural finding. Because semantically proximate posts share both topical content and audience behavior, the hashtag recommender surfaces tags that are topically relevant and empirically linked to similar engagement patterns. The system needs no retraining for new captions — a single SBERT forward pass, a K-nearest-neighbor lookup, and a frequency-weighted ranking delivers real-time recommendations.

1.11.2 Limitations

- **Non-textual confounders dominate.** Follower count, audio/music, thumbnail quality, posting time, and algorithmic amplification collectively explain far more engagement variance than caption text. The 60% classification ceiling directly reflects this.
- **Multilingual and vernacular gaps.** `all-MiniLM-L6-v2` is primarily English-trained. Non-English captions and TikTok-specific slang are embedded less reliably.
- **K sensitivity.** Variance reduction ranges from 10.5% to 13.0% depending on K and metric. There is no universally optimal K — the choice depends on the desired trade-off between neighborhood tightness and coverage.
- **Static corpus.** The KNN index is built on a fixed snapshot. New posts require index rebuilds; at scale, approximate nearest-neighbor structures (e.g., FAISS IVF) would be needed.

- **Selection bias.** The 3,976 posts dropped for having fewer than 4 clean characters (14.9%) skew the dataset toward posts with genuine descriptive content.

1.12 Part 8 — Hashtag Recommendation

1.12.1 8.1 Recommendation Algorithm

The combined extraction pipeline pushes hashtag coverage from ~4% (bridge table only) to 100% of the corpus. For each query, the system aggregates all hashtags from K neighbors and ranks them by (1) how often each hashtag appears among the neighbors, and (2) the average log-views of posts carrying that hashtag.

[26]:
```
def recommend_hashtags(
    query_idx: int,
    df: pd.DataFrame,
    knn_idx: np.ndarray,
    k: int,
    top_n: int = 10,
) -> pd.DataFrame:
    tag_stats = {}
    for nidx in knn_idx[query_idx]:
        row = df.iloc[nidx]
        eng = row["views_log"]
        for tag in row["hashtag_list"]:
            if tag not in tag_stats:
                tag_stats[tag] = {"count": 0, "total": 0.0}
            tag_stats[tag]["count"] += 1
            tag_stats[tag]["total"] += eng
    if not tag_stats:
        return pd.DataFrame(columns=["hashtag", "frequency", "avg_engagement"])
    recs = [
        {
            "hashtag":        t,
            "frequency":      s["count"],
            "avg_engagement": round(s["total"] / s["count"], 4),
        }
        for t, s in tag_stats.items()
    ]
    return (
        pd.DataFrame(recs)
        .sort_values(["frequency", "avg_engagement"], ascending=False)
        .head(top_n)
        .reset_index(drop=True)
    )


example_rec = recommend_hashtags(
    EXAMPLE_QUERY_IDX, df_processed, knn_indices[PRIMARY_K], PRIMARY_K
```

```
)
print(f"Query (clean): {df_processed.iloc[EXAMPLE_QUERY_IDX]['caption_clean']}")
print(f"Query hashtags: {df_processed.
 ↪iloc[EXAMPLE_QUERY_IDX]['hashtag_list']}\n")
print("Top-10 Recommended Hashtags:")
example_rec
```

```
Query (clean): Hip Movement Routine
Query hashtags: ['#functionalmovement', '#gymtok', '#fitnessforall',
'#hipmobiility', '#prenatalfitness']

Top-10 Recommended Hashtags:
```

[26]:

	hashtag	frequency	avg_engagement
0	#fitness	2	14.4186
1	#womenfitness	2	12.6711
2	#fyp	2	10.3352
3	#foryou	2	10.3352
4	#workoutchallenge	2	10.3207
5	#legday	2	9.8418
6	#legworkout	2	9.8418
7	#techtok	1	15.0683
8	#rave	1	15.0683
9	#ravetok	1	15.0683

1.12.2 8.2 Hashtag Coverage Improvement Report

[27]:
```
field_only_cov = df_processed["hashtags"].apply(
    lambda x: len(extract_hashtag_list(str(x))) > 0
).sum()
combined_cov = (df_processed["hashtag_list"].map(len) > 0).sum()
n = len(df_processed)

print("Hashtag Coverage Comparison")
print("-" * 50)
print(f"  Bridge table field only    : {field_only_cov:,} / {n:,}␣
 ↪({100*field_only_cov/n:.1f}%)")
print(f"  Combined (caption + field) : {combined_cov:,}   / {n:,}␣
 ↪({100*combined_cov/n:.1f}%)")
print(f"  Improvement                : +{combined_cov - field_only_cov:,}␣
 ↪posts")

avg = df_processed["hashtag_list"].map(len)
print(f"\n  Avg hashtags / post        : {avg.mean():.2f}")
print(f"  Median hashtags / post     : {avg.median():.0f}")
print(f"  Max hashtags / post        : {avg.max()}")
```

```
Hashtag Coverage Comparison
```

```
--------------------------------------------------
  Bridge table field only    : 883 / 22,647 (3.9%)
  Combined (caption + field) : 22,647   / 22,647 (100.0%)
  Improvement                : +21,764 posts

  Avg hashtags / post        : 5.58
  Median hashtags / post     : 5
  Max hashtags / post        : 138
```

1.12.3 8.3 Batch Recommendation Examples

Five queries with ground-truth hashtags, sampled for qualitative assessment of topical alignment.

[28]:
```python
def show_batch_recommendations(
    df: pd.DataFrame,
    knn_idx: np.ndarray,
    k: int,
    query_indices: list,
    top_n: int = 5,
) -> None:
    for qidx in query_indices:
        caption = df.iloc[qidx]["caption_clean"]
        actual  = df.iloc[qidx]["hashtag_list"]
        recs    = recommend_hashtags(qidx, df, knn_idx, k, top_n)
        print(f"\n{'' * 72}")
        print(f"Query [{qidx}]: {caption[:110]}")
        print(f"Actual tags : {actual[:8]}")
        if not recs.empty:
            print(f"Recommended (top {top_n}):")
            print(recs.to_string(index=False))
        else:
            print("No hashtags found in neighborhood.")


posts_with_tags = df_processed[df_processed["hashtag_list"].map(len) > 0]
SAMPLE_QUERIES  = list(posts_with_tags.sample(5, random_state=RANDOM_SEED).
 ↪index)

show_batch_recommendations(
    df_processed, knn_indices[PRIMARY_K], PRIMARY_K, SAMPLE_QUERIES, top_n=5
)
```

```
Query [14610]: We Are So Cooked is a funny and chaotic game coming this year!
indie games, new games, steam games, co-op game
Actual tags : ['#gaming', '#wearesocooked']
Recommended (top 5):
```

```
  hashtag  frequency  avg_engagement
  #gaming          4         12.8006
     #fyp          3         13.9927
   #games          3         11.8553
#fortnite          2         15.7755
     #ps5          2         14.2625


Query [8149]: $3 Dollar Tree DIY
Actual tags : ['#diy', '#dollartree', '#craft', '#wood', '#hack', '#homedecor',
'#craftgrooms']
Recommended (top 5):
         hashtag  frequency  avg_engagement
     #dollartree          9         13.0269
            #diy          7         13.2573
#dollartreefinds          6         12.6429
#dollartreehacks          5         13.0156
  #dollartreediy          5         11.7346


Query [18190]: Who been on ya mind
Actual tags : ['#lyricsvideo', '#lyrics_songs', '#pop', '#music']
Recommended (top 5):
    hashtag  frequency  avg_engagement
     #music          5         11.1863
       #fyp          5         10.3454
#applemusic          2          8.7691
  #acoustic          1         16.6487
   #singing          1         16.6487


Query [5513]: THIS HAD ME
Actual tags : ['#fyp', '#songwriter', '#improv', '#originalmusic', '#singer']
Recommended (top 5):
    hashtag  frequency  avg_engagement
       #fyp          5         14.0783
      #fyp          3         12.3522
     #funny          2         14.7114
#foryoupage          2         14.0314
    #foryou          2         12.3780


Query [16603]: How to create video game content only using an iPhone and PS5
Actual tags : ['#fyp', '#viral', '#eyegameuwatch', '#gaming', '#videogames',
'#tiktokgaming', '#inspiration', '#successmindset']
Recommended (top 5):
   hashtag  frequency  avg_engagement
    #viral          4         12.0664
```

```
      #fyp          4          11.1315
    #games          2          13.8593
#bestgames          2          13.8593
      #ps5          2          13.6665
```

1.13 Part 9 — Results Overview

The pipeline processed 26,623 raw posts down to 22,647 after removing captions with fewer than four non-hashtag characters (3,976 dropped, 14.9%). Binary engagement labels were assigned via median split on log-transformed view counts (median = 13.06), giving a near-perfectly balanced distribution: 11,324 high and 11,323 low engagement posts.

Classification baseline: All four models (Logistic Regression on SBERT, Multinomial Naïve Bayes on TF-IDF, KNN k=5 and k=10 on SBERT) converged to 59–60% accuracy and macro F1 of 0.58–0.59 on the 4,530-instance test set. The fact that dense and sparse representations converge at the same ceiling confirms this is an information limit, not a modeling problem.

Structural analysis: All 12 Mann-Whitney U tests (4 metrics × 3 K values) returned p 0.0. Variance reductions range from 10.46% (views, K=20) to 13.04% (likes, K=5) and decrease monotonically with K — as expected, since larger neighbourhoods include more distant posts and dilute semantic coherence.

K	likes_log	comments_log	views_log	shares_log
5	13.04%	12.72%	12.32%	12.46%
10	12.30%	12.74%	11.36%	11.98%
20	11.34%	11.27%	10.46%	10.53%

Hashtag coverage went from **3.9%** (883/22,647 from bridge table alone) to **100%** after extracting hashtags from caption text. Average hashtags per post: **5.58**. This was the most impactful single pipeline fix and the prerequisite for effective hashtag recommendation.

Recommendation: Batch examples show topically coherent recommendations for descriptive captions. Generic captions surface corpus-dominant tags, as expected from frequency-first ranking in sparse embedding regions.

1.13.1 9.2 Deployment — Online Inference

The structural finding is only useful if it can be put into practice. This section closes the loop: a new caption arrives, the system encodes it, retrieves its nearest neighbors from the corpus, and returns a ranked list of hashtags — all in a single function call.

The design is intentionally stateless — no retraining, no fine-tuning, no database writes. The `recommend_for_new_caption()` function accepts a raw caption string and produces recommendations by locating where that caption would sit in the learned embedding space. In practical terms, it tells a creator: *"posts that sound like yours tend to use these tags, and those tags are associated with this level of engagement"* — which is both actionable and transparent.

The three deployment examples below (fitness, cryptocurrency, meal prep) show that the system generalises across content verticals. Each query returns topically coherent, engagement-weighted recommendations drawn from similar posts in the corpus. Every suggestion traces back to real posts — no hallucination, no generic fallbacks.

Productionization paths: - **Streamlit app** — wrap `recommend_for_new_caption()` in a text input widget for instant creator-facing deployment - **FastAPI endpoint** — expose as a REST API; pre-load the model and corpus embeddings at startup for sub-100ms inference - **Batch offline** — run overnight over a refreshed corpus to pre-populate recommendation caches

1.13.2 9.1 Online Inference Function

`recommend_for_new_caption` takes a raw caption string and four context objects that are already in memory after running Parts 1–8:

Parameter	Type	Source
`raw_caption`	`str`	new user input
`model`	`SentenceTransformer`	loaded from `SBERT_MODEL_NAME`
`corpus_embeddings`	`np.ndarray` (N×384)	`embeddings` from Part 4
`df`	`pd.DataFrame`	`df_processed` from Part 3

The function is kept **standalone** rather than refactoring `recommend_hashtags` with an override parameter, for two reasons: 1. `recommend_hashtags` indexes into `df` via `query_idx` — a new caption has no row in `df`. 2. Keeping deployment logic isolated makes it straightforward to swap the embedding model or search index without touching the analysis code in Parts 5–8.

[29]:
```
def recommend_for_new_caption(
    raw_caption: str,
    model,
    corpus_embeddings: np.ndarray,
    df: pd.DataFrame,
    k: int = PRIMARY_K,
    top_n: int = 10,
) -> pd.DataFrame:
    clean_cap = build_caption_clean(raw_caption)
    query_vec = model.encode(
        [clean_cap], normalize_embeddings=True, convert_to_numpy=True
    )
    nn = NearestNeighbors(
        n_neighbors=k, metric="cosine", algorithm="brute", n_jobs=-1
    )
    nn.fit(corpus_embeddings)
    _, neighbor_idxs = nn.kneighbors(query_vec)
    neighbor_idxs = neighbor_idxs[0]

    tag_stats = {}
    for nidx in neighbor_idxs:
        row = df.iloc[nidx]
        eng = row["views_log"]
        for tag in row["hashtag_list"]:
            if tag not in tag_stats:
```

```
                tag_stats[tag] = {"count": 0, "total": 0.0}
            tag_stats[tag]["count"] += 1
            tag_stats[tag]["total"] += eng

    if not tag_stats:
        return pd.DataFrame(columns=["hashtag", "frequency", "avg_views_log"])

    recs = [
        {
            "hashtag":       t,
            "frequency":     s["count"],
            "avg_views_log": round(s["total"] / s["count"], 4),
        }
        for t, s in tag_stats.items()
    ]
    return (
        pd.DataFrame(recs)
        .sort_values(["frequency", "avg_views_log"], ascending=False)
        .head(top_n)
        .reset_index(drop=True)
    )
```

[30]:
```
_sbert_deploy = SentenceTransformer(SBERT_MODEL_NAME)

_demo_captions = [
    "Just hit a new personal record at the gym! Hard work always pays off",
    "Crypto market update - here is my honest take on what is happening right␣
 ↪now",
    "Healthy high-protein meal prep for the week, my go-to lunch routine",
]

print("=" * 72)
print("DEPLOYMENT DEMO - recommend_for_new_caption()")
print("=" * 72)

for _cap in _demo_captions:
    _recs = recommend_for_new_caption(
        _cap, _sbert_deploy, embeddings, df_processed, k=PRIMARY_K, top_n=8
    )
    print(f"\nCaption : {_cap}")
    print(f"Clean   : {build_caption_clean(_cap)}")
    if _recs.empty:
        print("  No recommendations (sparse neighborhood).")
    else:
        print(_recs.to_string(index=False))
    print("-" * 72)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
```

```
HF_TOKEN to enable higher rate limits and faster downloads.

Loading weights:   0%|          | 0/103 [00:00<?, ?it/s]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

======================================================================
DEPLOYMENT DEMO - recommend_for_new_caption()
======================================================================

Caption : Just hit a new personal record at the gym! Hard work always pays off
Clean   : Just hit a new personal record at the gym! Hard work always pays off
     hashtag  frequency  avg_views_log
    #fitness          7        10.1954
        #fyp          7         9.8922
     #gymtok          4        12.0038
        #gym          3         9.4739
      #viral          1        15.4642
     #foryou          1        15.4642
 #motivation          1        14.2210
#consistency          1        14.2210
----------------------------------------------------------------------

Caption : Crypto market update - here is my honest take on what is happening
right now
Clean   : Crypto market update - here is my honest take on what is happening
right now
        hashtag  frequency  avg_views_log
   #stockmarket          6         9.4498
#cryptocurrency          4        11.8414
        #stocks          4         9.7763
      #memecoin          3        11.9042
        #crypto          3        11.1499
       #finance          3         9.9816
           #fyp          2        11.8458
       #bitcoin          2        10.8898
----------------------------------------------------------------------

Caption : Healthy high-protein meal prep for the week, my go-to lunch routine
Clean   : Healthy high-protein meal prep for the week, my go-to lunch routine
          hashtag  frequency  avg_views_log
     #highprotein          3        13.0720
```

```
   #mealprepideas          2        12.9908
#highproteinmeals          2        11.4955
         #protein          2         9.2543
         #chicken          1        16.5628
           #bacon          1        16.5628
           #pasta          1        16.5628
            #food          1        16.5628
-----------------------------------------------------------------------
```